

The empirical recurrence for OEIS A283382, recovered from its tail

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8 September 2026

Abstract

OEIS A283382 records “Empirical recurrence of order 57” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 168$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 59$. Its last coefficient is $c_{57} = 3518743761 \neq 0$, so no further tail satisfies anything shorter; any order-57 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A283382 is “Number of $n \times 4$ 0..1 arrays with no 1 equal to more than two of its horizontal, vertical and antidiagonal neighbors, with the exception of exactly two elements.”. It has offset 1 and begins

0, 19, 532, 8087, 116624, 1592250, 20788531, 264040297, ...

The entry states:

Empirical recurrence of order 57 (see link above)

As of the “Last modified” line on the live entry (Mar 06 2017 17:05:04, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 168$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 168$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 57: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 57.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 336$ exact terms from $n = 2$ on, it returns order 57 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{57} c_i a(n-i),$$

c_1	24	c_{16}	−218050692	c_{31}	80515449032949	c_{46}	−1012115671617690
c_2	−126	c_{17}	806324043	c_{32}	−151975725019503	c_{47}	629006869901244
c_3	−406	c_{18}	−2413513359	c_{33}	275092811583769	c_{48}	−362807937369694
c_4	291	c_{19}	5278662960	c_{34}	−475789302081066	c_{49}	194744062107702
c_5	18633	c_{20}	−12944873334	c_{35}	782184141359502	c_{50}	−95550259884822
c_6	48787	c_{21}	33473943246	c_{36}	−1214080772463473	c_{51}	42088109836428
c_7	54645	c_{22}	−79421760405	c_{37}	1762686058738464	c_{52}	−16797092790261
c_8	−321570	c_{23}	184601932389	c_{38}	−2367055740337626	c_{53}	6083502330915
c_9	−339287	c_{24}	−428143900049	c_{39}	2911472681621595	c_{54}	−1858022049493
c_{10}	4521	c_{25}	963388998318	c_{40}	−3260656146433143	c_{55}	415852304049
c_{11}	5732622	c_{26}	−2129969522007	c_{41}	3318150761219037	c_{56}	−57139679259
c_{12}	−3550285	c_{27}	4649159443254	c_{42}	−3074875020588461	c_{57}	3518743761
c_{13}	1824138	c_{28}	−9867978302589	c_{43}	2613238026469971		
c_{14}	−64635282	c_{29}	20364054731136	c_{44}	−2055183336467556		
c_{15}	146834417	c_{30}	−41080295063324	c_{45}	1500520750988136		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 59$, and it is the recurrence OEIS A283382 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{57} - c_1 x^{57-1} - \dots - c_{57}$. Its constant term is $-c_{57} = -3518743761$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 57 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 57, so $\deg q = 57$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 57, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 57, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 3518743761.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-57 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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